Week 5 independent study

Geometry of $A\mathbf{x} = \mathbf{b}$ and Gaussian elimination

15 problems, increasing difficulty. Solutions intentionally omitted.

Independent practice 1: Row and column pictures

Solve $\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$. Describe both geometric pictures.

Show every elimination step and check your answer.

Independent practice 2: Three outcomes

Classify $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ c \end{bmatrix}$ for $c = 4$ and $c = 5$.

Show every elimination step and check your answer.

Independent practice 3: Find a multiplier

Eliminate below the first pivot in $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 8 & 13 & 29 \end{array}\right]$. Identify both pivots.

Show every elimination step and check your answer.

Independent practice 4: Back substitution

Solve $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 8 \end{bmatrix}$.

Show every elimination step and check your answer.

Independent practice 5: Full elimination

Solve $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \\ 1 & 2 & 3 & 14 \end{array}\right]$ by elimination.

Show every elimination step and check your answer.

Independent practice 6: Identify the elementary matrix

Find E implementing $R_3 \leftarrow R_3 - 4R_1$ on a 3×3 matrix.

Show every elimination step and check your answer.

Independent practice 7: Multiply EA

Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$. Compute EA and explain the operation.

Show every elimination step and check your answer.

Independent practice 8: The transformed right side

For $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 17 \end{bmatrix}$, use $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ to solve.

Show every elimination step and check your answer.

Independent practice 9: A zero pivot

Solve $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ by a row exchange. Find P.

Show every elimination step and check your answer.

Independent practice 10: A three-row permutation

Find P that exchanges rows 1 and 3 of $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$. Compute PA.

Show every elimination step and check your answer.

Independent practice 11: Singular but consistent

Solve $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$.

Show every elimination step and check your answer.

Independent practice 12: Singular and inconsistent

Change the previous right side to $\mathbf{b} = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$. Is the system solvable?

Show every elimination step and check your answer.

Independent practice 13: Column interpretation of consistency

For $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, which vectors $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ permit $A\mathbf{x} = \mathbf{b}$?

Show every elimination step and check your answer.

Independent practice 14: Order of matrix operations

Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ and $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Compare EP and PE.

Show every elimination step and check your answer.

Independent practice 15: Challenge: design a system

Construct a 3×3 system that needs a row swap, has three pivots, and has solution $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Show every elimination step and check your answer.